Figure Compilation: Walnut Patina Study

Figure 1

Correlation Between Excremental Viscosity Classification and Walnut Patina Formation Rate
(排泄物黏度等级与文玩核桃包浆形成速率的相关性)

Figure 1 illustrates the temporal evolution of walnut surface chromatic depth (ΔE value) under different biofluid viscosity regimes verified prior to immersion experiments.

Three experimentally validated viscosity classes were investigated:
- Stringy Medium (SM)
- Semi-solidified Matrix (SSM)
- High-Entropy Residue (HER)

Walnuts treated under HER conditions demonstrated significantly accelerated patina formation compared to conventional hand-polishing controls.

Patina formation rate Pr was approximated as:
$Pr = k \cdot \eta^{\alpha} \cdot M^{\beta}$

Where η represents viscosity index, M microbial activity coefficient, and k the psychological tolerance constant.

Figure 1 | Increasing viscosity and compositional entropy of biological treatment media significantly enhance walnut patina kinetics.

Figure 2

Comparative Evaluation of Walnut Patina Development Under Different Treatment Paradigms
(不同处理方式下文玩核桃包浆形成对比)

(a) Initial Condition (Control Baseline)

Figure 2A—初始状态（对照基线）

刚收到的核桃，仅进行了数天的人工接触处理。

表面仍较浅色，果肉残留未完全脱除，代表传统盘玩中常见的初始状态。

(b) Conventional Hand-Polishing Method

Figure 2B-传统手盘方法

经过数月持续人工盘玩的核桃。

尽管经历了长时间的人体接触，其颜色变化仍然缓慢且有限。

该结果说明：

| 传统手盘方式效率低、时间成本极高。

（c）Excremental Immersion Treatment (Experimental Group)

Figure 2C-排泄物浸泡实验组核桃被置于经过黏度核验的高熵生物介质中浸泡。

样品表现出：

• 快速变色

• 表面光泽提升

•在极短时间内形成明显

结果从视觉上支持：

| 富含氮的生物环境能够显著加速“老化

Figure 2 | Visual comparison of walnut patina progression under conventional manual polishing and excrement-assisted immersion treatment. While traditional methods require months to years of handling, experimental biofluid exposure achieves comparable or enhanced maturation within significantly shortened durations.

Supplementary Materials

S1. Odor Persistence Test

To evaluate post-treatment usability, odor persistence was monitored across all experimental groups under ambient laboratory conditions. Samples exposed to high-entropy biological media demonstrated prolonged olfactory retention exceeding 72 hours.

Olfactory intensity was assessed using a semi-quantitative researcher response scale (0 = undetectable, 5 = immediate evacuation response). HER samples consistently scored above 4.5, indicating significant sensory impact.

S2. Researcher Exposure Limitation Protocol

Due to increasing psychological and environmental stress observed during experimental handling, a controlled exposure protocol was implemented. Researchers were limited to a maximum of 5-minute interaction intervals followed by mandatory ventilation periods.

Personal protective equipment (PPE) included gloves, masks, and optional existential distancing.

S3. Biofluid Renewal Cycle

To maintain consistent biochemical activity, biological media were periodically renewed according to a standardized replacement schedule. Renewal frequency was determined based on visual turbidity, viscosity deviation, and laboratory morale decline.

The renewal cycle ensured sustained nitrogen availability and microbial activity, supporting continuous patina acceleration.

Supplementary Note | These materials provide methodological transparency for the proposed bio-assisted walnut maturation framework and address reproducibility considerations.